

When Does Multimodal Diffusion Inverse Kinematics Help Soft Continuum Arms—and Does It Survive Model Mismatch?

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Abstract—Soft continuum arms are highly redundant: many actuation patterns realize the same tip position. Deterministic learned inverse kinematics (IK) regressors average these incompatible solutions into invalid ones, while a conditional diffusion model can represent the full solution distribution. We present a controlled simulation study that asks two questions the existing literature leaves open: *when* does this multimodality actually help, and does the advantage survive a change of dynamics model? On a piecewise-constant-curvature (PCC) pneumatic arm, a conditional diffusion model matches a deterministic baseline where redundancy is absent and dominates where it is present (mode recall 0.84 ± 0.03 vs. exactly 0; obstacle-avoidance success 97.8% vs. 4.0%; three seeds), and matches a strong optimization-based solver’s task success, mode recall, and solution diversity at a $5\text{--}11\times$ lower query cost, without needing the simulator at query time. Two controlled manipulations show the advantage is specifically a redundancy effect: conditioning on the whole arm shape instead of the tip collapses sampled diversity by $\sim 80\times$ and shrinks the baseline’s disadvantage from $\sim 30\times$ to $\sim 3\times$, while increasing the number of segments widens the gap. However, when actuations from the PCC-trained model are executed on a high-fidelity Cosserat-rod simulator, accuracy collapses on every seed ($0.67\text{ mm} \rightarrow 139.9\text{ mm}$, 0% success): the learned inverse map is simulator-specific. We characterize what recovers this loss: candidate selection under a query budget helps but is bounded; a diversity-aware query order significantly outperforms a confidence order (paired $t=3.3$); guidance-based adaptation yields a modest, statistically suggestive gain; gain-domain-randomization fails entirely; and fine-tuning on target-domain data reveals a sharp retention–transfer cliff with no data-mixing sweet spot. All code, data, and per-experiment result files are made available.

Index Terms—Modeling, Control, and Learning for Soft Robots; Deep Learning Methods; inverse kinematics; sim-to-sim transfer.

I. INTRODUCTION

A MULTI-SEGMENT pneumatic continuum arm with three chambers per segment has more actuation degrees of freedom than task degrees of freedom: for a four-segment arm, a 12-D actuation vector determines a 3-D tip position. The inverse map is therefore one-to-many, and its preimages are not merely a connected null space—distinct arm shapes reach the same tip. A deterministic learned IK regressor trained by motor babbling [1] fits the conditional *mean* of this multimodal distribution and can return an actuation that realizes none of the true solutions. A conditional diffusion

model [2], [3] instead learns the full conditional distribution $p(\mathbf{q} \mid \text{target})$ and can sample all solution modes.

Diffusion models for robot action generation are now well-established [4], and diffusion-based IK has been demonstrated for rigid multi-arm systems [5]. What the literature does not yet provide for *soft continuum* arms is (i) a controlled account of when the multimodality actually matters—as opposed to diffusion simply being a strong regressor—and (ii) an honest measurement of whether the learned inverse map survives the model mismatch that any simulation-trained soft-arm controller must eventually face. Soft arms make question (ii) unusually pressing: their dynamics are hard to model, so the gap between any two models of the *same* physical arm is large, and a method whose advantage evaporates under that gap is of limited practical value.

We answer both questions in a fully reproducible sim-to-sim setting: an idealized, fast PCC model [6] as the training simulator, and a Cosserat-rod simulation (PyElastica [7], [8]) of the same arm as the high-fidelity evaluation domain. Our contributions:

- 1) **A validated advantage.** With three-seed statistics, diffusion IK dominates learned MLP and mixture-density-network (MDN) baselines, recovers 0.84 ± 0.03 of ground-truth IK modes where both recover none, and matches a classical random-restart optimization solver on task success, mode recovery, and solution diversity at a $5\text{--}11\times$ lower query cost (single-target vs. batched) without simulator access at query time—translating into a 97.8% vs. 4.0% obstacle-avoidance gap over the learned baselines (Sec. IV).
- 2) **The mechanism, isolated.** Two controlled manipulations of redundancy—conditioning on the whole body shape, and varying the number of segments—show the advantage is specifically redundancy resolution, not generic model capacity (Sec. V).
- 3) **A quantified transfer collapse—and what recovers it.** The learned inverse map collapses under simulator change on every seed. We quantify the recovery achievable by inference-time selection and guidance, show a diversity-aware query ordering significantly outperforms the obvious confidence ordering (paired $t=3.3$, $p \approx 0.002$), show gain-domain-randomization achieves nothing, and map a sharp retention–transfer cliff for target-domain fine-tuning (Sec. VI).

Negative and bounded results are reported with the same

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prominence as positive ones; Sec. VII states the limitations explicitly.

II. RELATED WORK

IK for continuum arms. Non-learned approaches solve continuum IK by optimization through a model at query time: Jacobian-based schemes for nonconstant-curvature arms [1] and real-time FEM inversion [9]; the constant-curvature kinematics underlying our fast simulator are reviewed in [6], and STIFF-FLOP-style pneumatic arms motivate the three-chamber actuation model [10]. We include an optimization-based baseline of this family (random-restart gradient IK through the differentiable model). Learned approaches fit a deterministic target-to-actuation map on motor-babbling data [1]; mixture density networks [11] are the classical answer to multimodal regression and serve as the learned multimodal baseline.

Diffusion models for robots. Denoising diffusion models [2] with DDIM sampling [3] and classifier-free guidance [12] underpin diffusion policies [4]. IKDiffuser [5] solves IK for *rigid multi-arm* systems with flexible kinematic-tree *topology*; physically compliant continuum bodies are outside its scope. Selecting one action from a diffusion policy’s samples via learned scorers is an active area; our transfer-guided sampling instead injects the score’s gradient *during* generation, in the spirit of classifier guidance [13], and our query-budget framing asks how many *expensive target-domain evaluations* selection needs, which we have not seen posed for soft-arm IK.

Crossing the reality gap. Domain randomization [14] and target-domain fine-tuning are the standard levers; catastrophic forgetting under naive fine-tuning is well-documented [15]. We evaluate both levers in a controlled sim-to-sim instance and find the first fails completely while the second exposes a sharp retention–transfer tension.

III. SETUP

Arm and training simulator. The training domain is a four-segment PCC arm (segment length 60 mm, total length 240 mm). Each segment has three 120°-spaced chambers with normalized pressures $\mathbf{q} \in [0, 1]^{12}$; a linear chamber-to-curvature map (gain 25) yields per-segment curvature components (k_x, k_y) , and forward kinematics composes constant-curvature arcs. The model is batched and differentiable.

Data and models. Training data are 5×10^5 i.i.d. uniform actuations pushed through forward kinematics. The diffusion model is a residual MLP denoiser (six FiLM-conditioned blocks, width 512) trained as a DDPM ($T=1000$, cosine schedule) with classifier-free guidance (condition dropout 0.1); sampling uses DDIM. Ablations (Sec. IV) fix guidance weight 1.5 and 20 DDIM steps. Baselines: an MLP regressor and an MDN (eight mixture components), matched in width, plus a classical optimization-based reference—gradient-descent IK through the differentiable PCC model from 32 random restarts (Adam, 300 iterations), denoted grad-IK. Unless stated otherwise we sample $K=32$ candidates per target and report best-of- K ; success means tip error under 5 mm (2% of arm length).

Accuracy (E1) is evaluated on 512 held-out targets and mode recall (E2) on 20. The obstacle task (E3) runs 200 trials, each placing three random spherical obstacles (radius 2 cm) in the workspace; success requires any of the K samples to be simultaneously accurate and collision-free.

Ground-truth modes. For evaluation targets we enumerate the true IK solution set by a 2×10^6 -sample sweep, keeping actuations whose tip lands within tolerance and clustering with DBSCAN [16] ($\varepsilon=12$, $\text{min_samples}=5$; a generated sample matches a mode within radius 15). Clustering, mode matching, and diversity are computed in the 8-D per-segment *curvature* space (k_x, k_y) rather than raw pressures: the chamber-to-curvature map has a per-segment null space (common-mode pressure offsets change $\mathbf{q}$ but not the shape), which would otherwise contaminate distance-based metrics. Evaluation targets average 2.05 ground-truth modes (range 0–6); targets where the sweep finds no mode are excluded from recall averages.

High-fidelity domain. The transfer domain is a Cosserat-rod simulation of the same arm in PyElastica [7], [8]: one rod, 48 elements, per-segment constant internal couples derived from the same chamber pressures, integrated with linear damping from rest for a fixed 1.5 s horizon (~ 4 s of compute per evaluation). A convergence check (eight random actuations) shows the damped rod still oscillates at this horizon and beyond (tip shifts of 50–145 mm persist between horizons up to 9 s), so the transfer domain is a *fixed, deterministic dynamic-snapshot protocol*—applied identically in every Elastica experiment, dataset, and label in this paper—rather than a static equilibrium. The two simulators agree exactly in the zero-curvature case—uniform pressures produce the identical straight configuration (tip at $[0, 0, 0.24]$ m) in both—so the discrepancy lies in the bending response, not in trivial misalignment. Sweeps over the couple gain and the damping constant further confirmed the discrepancy is directional/structural rather than a calibration artifact: no gain spanning two orders of magnitude and no damping in 0.5–10 aligns the two workspaces (mean tip discrepancy never falls below ~ 150 mm on shared actuations, though the workspaces are similarly sized), and the transfer collapse of Sec. VI persists at every tested damping (110–145 mm best-of-8, 0% success).

IV. WHEN MULTIMODALITY HELPS

Table I summarizes the four core experiments. **E1 (accuracy):** best-of- K diffusion reaches 0.44 mm mean tip error with 100% success; the deterministic MLP, structurally limited to the conditional mean, plateaus at 13.4 mm, and the MDN does not improve on it. The advantage is not merely the $K=32$ shots: a *single* diffusion sample reaches 1.69 ± 0.15 mm—still $8\times$ better than the MLP—while a single MDN draw lands at 84 mm. The classical grad-IK reference reaches 0.013 mm at 100% success—when the true model is available to optimize through at query time, classical optimization is the accuracy ceiling. **E2 (multimodality):** against enumerated ground-truth solution sets, diffusion recovers 0.84 ± 0.03 of modes; both baselines recover none—the MLP because it is deterministic, and the MDN not through classic component collapse but

TABLE I

CORE COMPARISON ON THE PCC ARM (MEAN $\pm$ STD, 3 SEEDS; $K=32$). MODE RECALL AND DIVERSITY ARE COMPUTED OVER SUCCESSFUL SAMPLES ONLY. GRAD-IK IS A SINGLE OPTIMIZATION RUN PER TARGET (NO TRAINING, HENCE NO SEEDS; RESTART-RNG VARIANCE IS NEGLIGIBLE OVER THREE RNG SEEDS) AND REQUIRES THE DIFFERENTIABLE SIMULATOR AT QUERY TIME. LEARNED-METHOD QUERY TIMES AMORTIZE A 64-TARGET BATCH; A SINGLE-TARGET DIFFUSION QUERY COSTS ~ 84 MS.

	Diffusion	grad-IK	MLP	MDN
Tip err. best-of- K (mm)	0.44 $\pm$ 0.03	0.013	13.4 $\pm$ 0.0	17.3 $\pm$ 0.2
Tip err. single sample (mm)	1.69 $\pm$ 0.15	—	13.4 $\pm$ 0.0	83.6 $\pm$ 2.7
Success @ 5 mm	1.00	1.00	0.060	0.027
Mode recall	0.84 $\pm$ 0.03	0.85	0.00	0.00
Diversity (curv. space)	28.2 $\pm$ 0.2	29.0	0.0	0.0
Obstacle success	0.978 $\pm$ 0.002	0.980	0.040	0.018
Query time (ms/target)	38.6 $\pm$ 4.9	428	0.013	0.022

because no component is accurate enough: mode recall and diversity are computed over *successful* samples only, and the MDN almost never produces one. Per-component analysis rules out a training pathology—NLL decreases monotonically at every epoch, no variance reaches its clamp floor, and 7.8 of 8 components remain effective—yet even the *best* component mean per target carries 7.5 mm error (median component 9.2 mm) against the 5 mm tolerance, and sampling with the learned σ smears this further. Nor is it a tuning artifact: sweeping the component count over $\{4, 8, 16, 32\}$ improves best-of- K error only from 18.7 to 14.8 mm—never surpassing the plain MLP—and mode recall remains exactly zero at every setting. grad-IK’s random restarts, by contrast, *do* recover modes (recall 0.85) with solution diversity on par with diffusion’s (29.0 vs. 28.2)—multimodality requires representing multiple candidates, not learning; the learned baselines fail because they cannot. These conclusions are not artifacts of the enumeration hyperparameters: across a 3×3 sensitivity grid ($\varepsilon \in \{10, 12, 14\}$, match radius $\in \{10, 15, 20\}$), absolute recall shifts with the radius—which *defines* mode recovery—but the learned baselines stay at zero everywhere, while diffusion and grad-IK trade the lead (diffusion higher at three settings, grad-IK at five, one tie): the two multimodal methods are on par at every setting of the enumeration. **E3 (downstream value):** when success requires *any* of K samples to be both accurate and collision-free among random obstacles, diffusion succeeds on 97.8% of trials vs. 4.0% (MLP)—redundancy is only exploitable if the model can represent it. **E4 (cost):** 38.6 ms per target (batch of 32 samples, CPU, 64 targets batched; ~ 84 ms for a single-target query) versus 428 ms for grad-IK—a 5–11 $\times$ gap. A tuned budget narrows it: grad-IK at 100 iterations (~ 130 ms) still reaches 0.30 mm at 100% success, so with the simulator in hand, classical optimization is cost-competitive within a small factor. The learned methods’ advantage is different in kind: they answer from input–output samples alone, with no differentiable simulator in the loop at query time. (One caveat applies to both: best-of- K reporting assumes a verifier to pick the successful candidate—here K forward evaluations, versus grad-IK’s $K \times 300$ forward–backward passes; in deployment that role falls to perception.) The amortization property is exactly what Sec. VI puts under stress. With three seeds, formal hypothesis testing adds little;

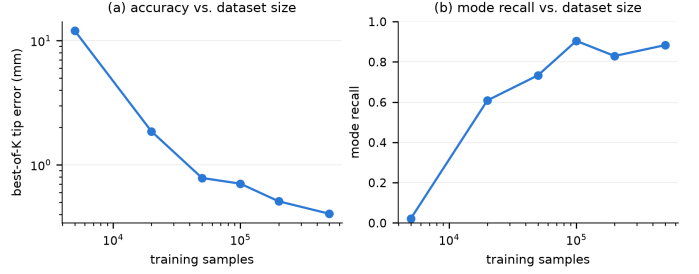


Fig. 1. Dataset-size ablation (single seed). (a) Best-of- K tip error and (b) mode recall vs. training-set size; both saturate by $\sim 0.5\text{--}1 \times 10^5$ samples.

the separations are instead unambiguous by inspection: on every task metric (accuracy, recall, diversity, obstacle success) the *worst* diffusion seed outperforms the *best* learned-baseline seed—for tip error, 0.49 vs. 13.33 mm, a gap exceeding 400 $\times$ the across-seed standard deviation. (Query time is excluded—the MLP is trivially fastest—as is grad-IK, which has no training seeds and wins raw accuracy.)

Three ablations support the protocol: guidance weight has an inverted-U optimum at 1.5 (both accuracy and diversity collapse beyond ~ 3); accuracy and recall saturate by ~ 20 DDIM steps (a 3 $\times$ speedup over 50 steps at no measurable cost); and performance saturates by $\sim 0.5\text{--}1 \times 10^5$ training samples (Fig. 1).

V. THE ADVANTAGE IS REDUNDANCY, SPECIFICALLY

If the advantage in Table I is redundancy resolution, it should shrink when redundancy is removed and grow when redundancy is added. Both predictions hold (Fig. 2).

Removing redundancy: shape conditioning. Conditioning the same models on the full 24-D body shape (eight backbone points) instead of the 3-D tip pins down almost the entire configuration. Diffusion’s sampled diversity collapses from 28.16 ± 0.22 to 0.346 ± 0.022 ($\sim 80\times$), and the MLP’s accuracy disadvantage shrinks from $\sim 30\times$ (13.4 vs. 0.44 mm tip error) to $\sim 3\times$ (0.49 vs. 0.16 mm mean per-backbone-point error, the shape task’s accuracy metric), with both models at 100% success (3 seeds). When there is one answer, a deterministic regressor is nearly as good; diffusion’s edge is specifically the multimodal case.

Adding redundancy: morphology. Training fresh models on two-, six-, eight-, and ten-segment variants of the same arm family (tolerance scaled to 2% of arm length), the MLP’s error grows monotonically from 1.7 mm (2 seg, 75% success) through 33.9 and 62.5 mm to 103.2 mm at ten segments, with 0% success from eight segments on, while diffusion degrades gracefully (0.11 to 3.2 mm, $\geq 99.8\%$ success throughout); from eight segments the obstacle-task separation is total ($\geq 99.5\%$ vs. 0%). More redundancy means worse mode averaging and more collision-free alternatives that only a multimodal model can offer.

VI. DOES IT SURVIVE MODEL MISMATCH?

A. The transfer collapse

Executing the PCC-trained diffusion model’s actuations on the Cosserat-rod simulator collapses accuracy from 0.67 $\pm$

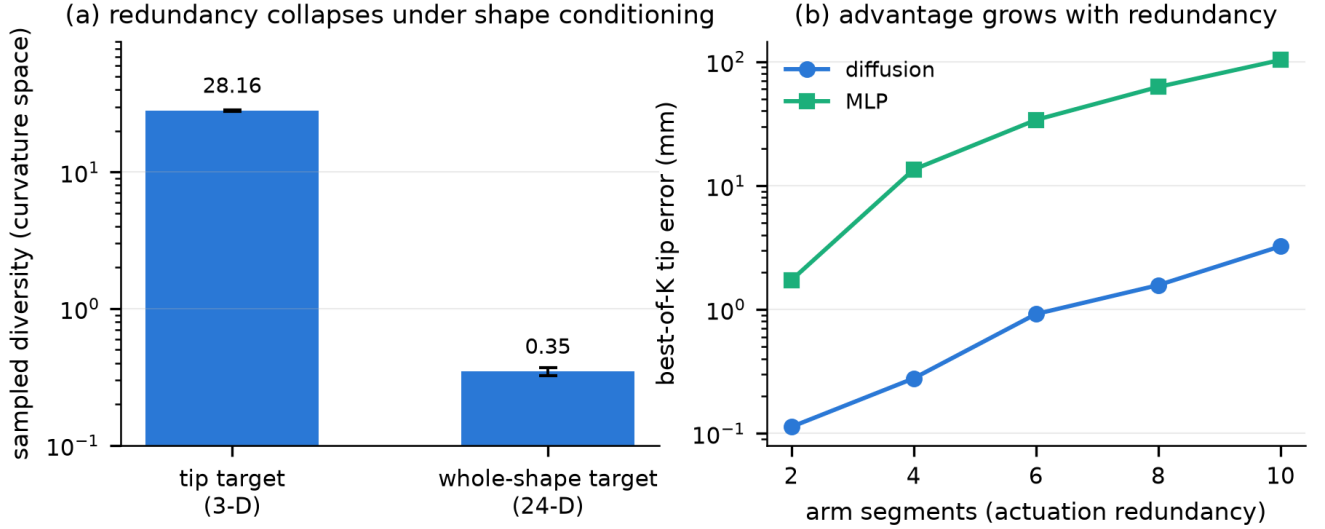


Fig. 2. The diffusion advantage tracks actuation redundancy. (a) Replacing the 3-D tip target with a 24-D whole-shape target removes most of the redundancy: sampled diversity collapses $\sim 80\times$ (28.16 ± 0.22 to 0.346 ± 0.022 ; mean $\pm$ std, 3 seeds). (b) Increasing the number of segments (single seed per morphology) widens the diffusion–MLP accuracy gap monotonically from two to ten segments; from eight segments on, the MLP’s obstacle-task success is 0% while diffusion’s stays $\geq 99.5\%$.

TABLE II

RECOVERY MECHANISMS AT A GLANCE (ELASTICA-DOMAIN PERFORMANCE; BEST CONFIGURATION OF EACH MECHANISM; EACH ROW USES ITS OWN STUDY’S HELD-OUT TARGET SET, 15–45 TARGETS). “QUERIES” ARE TARGET-DOMAIN (ELASTICA) EVALUATIONS PER TARGET; THE TOP-1 ROW SPENDS NONE. THE 139.9 MM TRANSFER-COLLAPSE FIGURE QUOTED ELSEWHERE CORRESPONDS TO BEST-OF-8 SELECTION AND IS CONSISTENT WITH THIS STUDY’S $m=8$ POINT (139.7 MM).

Mechanism	err (mm)	succ.	Cost / caveat
none (top-1 candidate)	161.5	0%	no target queries
best-of-32 selection	130.4	0%	32 queries/target
+ diversity order	–6.7 at $m=8$	0%	same queries; $p \approx 0.002$
guided sampling	129.0	0%	400 labels; suggestive
gain randomization	140.6	0%	degrades source; no gain
fine-tune, 0% mix	67.1 ± 0.9	10%	forgets source (133 mm)
fine-tune, $\geq 5\%$ mix	≥ 133.8	$\leq 10\%$	retains source, forfeits gain

0.03 mm (100% success) to 139.9 ± 1.8 mm with 0% success on every seed ($n=20$ held-out targets, $K=8$); the MLP collapses similarly ($13.8 \rightarrow 164.4$ mm). The learned inverse map is simulator-specific. The rest of this section quantifies what does—and does not—recover the loss; Table II summarizes.

B. Selection under a query budget

A multimodal model offers something a deterministic one cannot: K diverse candidates to test against the target domain. Spending a budget of m Elastica evaluations and keeping the best candidate reduces mean error from 161.5 ($m=1$) to 130.4 mm ($m=32$), improving all 45 of 45 targets (Fig. 3)—real and broad-based, but to be blunt about its practical value: success stays at 0%, the gap to tolerance remains $\sim 26\times$, and under this budget selection never actually solves a single target. How the budget is spent matters: a greedy farthest-point ordering in curvature space (seeded from the same most-confident candidate, so $m=1$ is identical) beats confidence ordering by 6.7 mm at $m=8$ (paired $t=3.34$, $p \approx 0.002$; 24/12

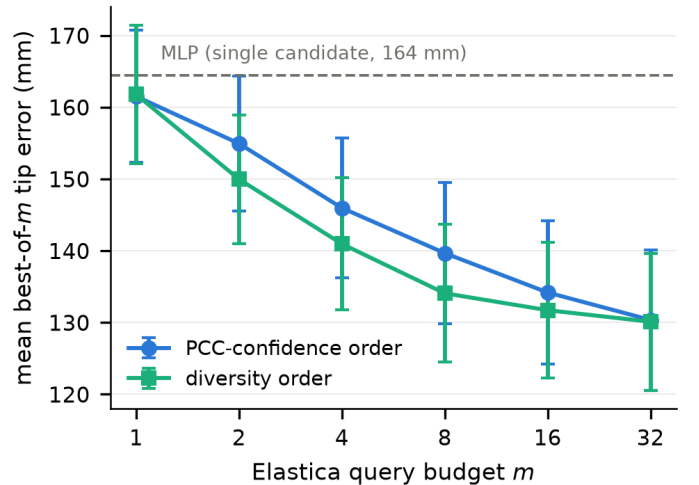


Fig. 3. Candidate selection under an Elastica query budget ($n=45$ targets, $K=32$ candidates, mean $\pm$ SEM). Curves shown are independent runs of the two orderings; the statistical comparison in the text uses a *paired* design (one candidate set per target, both orderings reindex the same cached per-candidate errors), which is significant at $m=8$ and $m=16$ ($t=3.34$ and 3.21 , $p \approx 0.002$).

targets) and 3.3 mm at $m=16$ ($t=3.21$). Notably, neither candidate diversity itself nor local PCC Jacobian sensitivity predicts *which* targets benefit: at $n=45$ —sized to detect $r \geq 0.41$ at 80% power, i.e., the effect size an $n=20$ pilot had suggested—the observed correlations are $r=0.20$ ($p=0.19$) and $r=0.02$ ($p=0.88$). This rules out the pilot’s effect but not smaller ones; the pilot’s non-replication is a caution we report rather than hide.

C. Guiding generation toward the target domain

Beyond selecting among candidates, we steer generation itself: a small regressor (two hidden layers, width 128) trained on 400 Elastica-labeled candidates (25 targets) predicts trans-

TABLE III

FINE-TUNING ON 10^4 ELASTICA-SIMULATED SAMPLES AT SIX PCC-DATA MIXING RATIOS (MEAN $\pm$ STD, 3 SEEDS). COMMON PROTOCOLS ACROSS ALL ROWS, BASELINE INCLUDED: PCC COLUMNS USE THE E1 EVALUATION (512 TARGETS, $K=32$); ELASTICA COLUMNS USE 20 HELD-OUT TARGETS, $K=8$. THIS TARGET DRAW IS HARDER THAN THE TRANSFER-COLLAPSE EXPERIMENT’S (151.1 VS. 139.9 MM AT BASELINE).

PCC mix	PCC err (mm)	succ.	Elastica err (mm)	succ.
none (baseline)	0.44 ± 0.03	100%	151.1 ± 2.5	0%
0%	133.4 ± 1.2	0%	67.1 ± 0.9	10%
5%	30.7 ± 0.6	6%	133.8 ± 3.1	8%
10%	21.7 ± 0.5	17%	143.2 ± 4.9	8%
15%	16.7 ± 0.2	26%	146.0 ± 3.4	8%
20%	11.6 ± 0.3	51%	148.0 ± 1.6	10%
50%	3.0 ± 0.4	96%	152.3 ± 0.4	3%

fer error from (candidate, target), and its gradient nudges the denoised estimate at every DDIM step, as in classifier guidance [13] (guidance scale 0.3, fixed a priori without tuning). On 15 held-out targets this improves best-of-16 error from 134.6 to 129.0 mm (paired 5.6 ± 2.5 mm, $t=2.27$, $p \approx 0.04$; 11/15 targets). This is generalization to unseen targets, but it is the weakest effect in the paper: all p -values here are uncorrected, and across the family of $\sim$ six hypothesis tests we report, only the ordering result ($p \approx 0.002$) survives a Bonferroni correction—we therefore read this one as *suggestive* rather than confirmed. Combining it with diversity-ordering adds nothing: guidance *concentrates* the candidate distribution while diversity-ordering *spreads* selection, and a 2×2 factorial shows the mechanisms in tension rather than compounding.

D. Training-time levers

Domain randomization fails. Randomizing the curvature gain per-sample ($\mathcal{U}(15, 35)$) during data generation costs PCC accuracy ($0.44 \rightarrow 2.18$ mm) and yields *zero* transfer improvement ($140.6 \rightarrow 140.6$ mm)—consistent with the mismatch being directional rather than magnitude-based: randomizing how much the arm bends cannot fix a discrepancy in which direction it bends.

Target-domain data works—at a price.

Fine-tuning the pretrained model *exclusively* on 10^4 Elastica-simulated samples (20 epochs, learning rate 3×10^{-5} ; mixing ratios below are realized in expectation by weighted sampling within each batch) yields the only nonzero transfer success of any method— 67.1 ± 0.9 mm and 10% success (2/20 targets on each seed; recall the 5% metric granularity)—at the cost of catastrophic forgetting [15] of the source domain ($0.44 \rightarrow 133.4$ mm). To be plain about magnitude: even this best transfer result remains $\sim 13 \times$ the success tolerance in mean error; no method in this paper solves the transferred task. Mixing PCC data back into fine-tuning batches restores retention smoothly and monotonically across all six ratios ($133.4 \rightarrow 30.7 \rightarrow 21.7 \rightarrow 16.7 \rightarrow 11.6 \rightarrow 3.0$ mm), but the transfer gain does not degrade in step: the fine sweep *locates* the collapse below the first grid point—at just 5% PCC data, transfer error has already rebounded from 67.1 to 133.8 ± 3.1 mm, forfeiting $\sim 80\%$ of the improvement over the same-target no-fine-tune baseline (151.1 ± 2.5 mm), and

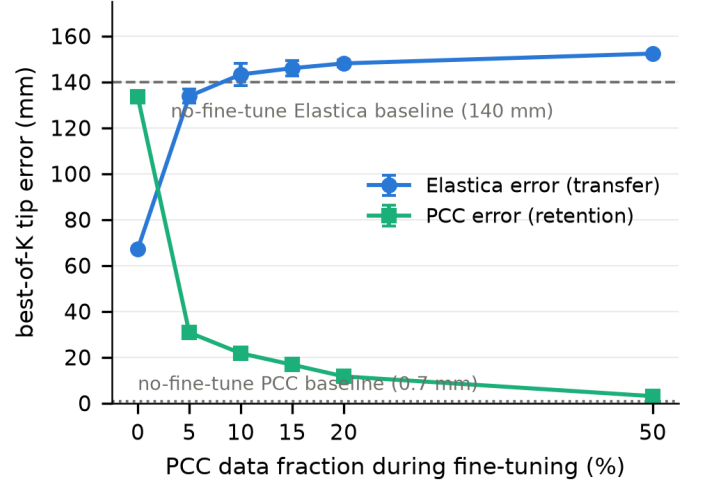


Fig. 4. The retention–transfer cliff (mean $\pm$ std, 3 seeds; six mixing ratios). Fine-tuning the PCC-pretrained model on 10^4 Elastica samples: retention (PCC error) improves smoothly and monotonically as PCC data is mixed back in, but the transfer gain collapses almost entirely by the first grid point (5% PCC data). There is no data-mixing sweet spot.

it drifts only slowly thereafter (Table III, Fig. 4). The values at intermediate ratios are stable across seeds (std ≤ 4.9 mm), so this is not an unlucky training run. Retention and transfer sit on a *cliff* whose edge lies below 5% mixing, not on a smooth tradeoff curve—under per-batch data mixing there is no sweet spot, suggesting parameter-level regularization (e.g. elastic weight consolidation [15]) as the necessary next mechanism. (Binary success has 5% granularity at $n_{\text{targets}}=20$; mean error carries the statistical weight throughout.)

VII. LIMITATIONS

(1) **Simulation-only.** Both domains are simulators; this is a controlled sim-to-sim study, and no claim is made about physical hardware. (2) **The target domain is a dynamic snapshot, not an equilibrium.** The Cosserat-rod arm applies idealized per-segment couples over a fixed 1.5 s damped-integration horizon at which the rod demonstrably has not settled (see Setup); the domain is a consistent, deterministic protocol and a harder test, but neither a static posture nor a validated model of a specific physical arm. (3) **Statistical coverage is uneven.** Core results, the shape-conditioning study, the baseline transfer collapse, and the fine-tuning grid carry three-seed statistics; the query-budget, ordering, and guidance experiments use one trained model (with within-experiment pairing across 15–45 targets), the hyperparameter, dataset-size, and morphology studies are likewise single-seed, and the grad-IK reference is a single optimization run per target. (4) **Mode-enumeration hyperparameters** were calibrated for the four-segment arm; morphology results therefore report accuracy and obstacle-task metrics only. (5) **Success-rate granularity** in transfer experiments is 5% ($n_{\text{targets}}=20$).

VIII. CONCLUSION

For the PCC-representable pneumatic continuum arms studied here, multimodal diffusion IK delivers a real, mechanistically attributable advantage—it is redundancy resolution,

growing with segment count and vanishing when the target specifies the whole shape—but the learned inverse map itself does not survive a change of dynamics model. Inference-time selection and guidance recover a bounded fraction of the loss; synthetic gain randomization recovers none; and target-domain fine-tuning buys genuine transfer only at a sharp retention cliff that data mixing cannot smooth. For simulation-trained soft-arm IK, the practical path forward is not better sampling from a source-domain model but mechanisms that let a small amount of target-domain data reshape the model without erasing it.

REPRODUCIBILITY

All code, configuration, per-experiment result files, and the 10^4 -sample Elastica dataset are available at <https://github.com/adhvayjagadeesh/soft-arm-diffusion-ik>.

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